



Effects of charge cut-off voltage on the performances of monocrySTALLINE $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ /graphite Li-ion cells

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ARTICLE INFO

Article history:

Received 22 October 2018

Received in revised form

28 January 2019

Accepted 28 January 2019

Available online 29 January 2019

Keywords:

MonocrySTALLINE $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$

Charge cut-off voltage

Capacity fading

Electrochemical performance

Electrode-electrolyte interfaces

ABSTRACT

Driven by the growing demands of electric vehicles (EVs) and hybrid electric vehicles (HEVs), high energy density Lithium-ion batteries (LIBs) have attracted extensive attentions. Enlarging the charge cut-off voltage (COV) is one of the most effective strategies to improve the energy density of LIBs. In this paper, the electrochemical performances of monocrySTALLINE $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (MNCM523)/artificial graphite (AGr) pouch Lithium-ion cells charged to 4.30 V, 4.35 V and 4.40 V are studied. The results show that with the increase in the COV, the electrochemical performances including capacity, median voltage and energy density of the cell are all improved with the increases of ohmic resistance simultaneously, while the rate capabilities, high and low temperature (HT<) discharge performances and storage performance become slightly worse. The cycle performance of the cells at different charge voltages has no significant difference within 800 cycles but deteriorates after 1000 cycles. The capacity retention gradually decreases with the increasing of COV. Electrochemical impedance spectroscopy (EIS) and incremental capacity analysis (ICA) are adopt to explain the possible reasons of capacity fading, which is considered caused by continuous loss of active materials and lithium inventory due to the deterioration of the electrode-electrolyte interfaces and the increase of charge transfer resistance. The results give a guidance for the design and application.

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1. Introduction

With the shortage of petroleum resources and the improvement of people's awareness of environmental protection, new energy vehicles are gradually replacing fuel vehicles and becoming important transportations. Lithium-ion batteries (LIBs) are the most suitable rechargeable batteries due to their excellent cycle life, relatively high energy density and favorable power performance. However, with the requirement of long-term travelling ability for vehicles, high energy density has become the most important factor because it determines the usage time after per charge and affects the drive distance for the electric vehicles [1,2]. As the

limited capacity of the mainly used commercial carbon-based anode materials, developing new cathode materials is extremely most important to realize high energy density LIBs. Among the various candidates of cathode materials, the layered nickel-cobalt-manganese oxide materials (referred as NCM) [3,4] which possesses a high specific capacity, long cycle life, good safety performance and relatively low cost, is considered to be an ideal candidate for high energy LIBs for electric vehicles comparing with LiCoO_2 , LiNiO_2 and LiMnO_2 [5–7]. The conventional NCM usually has a smooth spherical surface and is composed of aggregated sub-micrometer primary particles [8]. Great efforts have been focused on the modifications of the layered NCM such as structure building, element doping or coating [9–12]. However, this kind of structure itself has the disadvantages of low compaction density and specific capacity which usually fail to meet the needs of high energy density batteries. Besides, the demand of high energy density batteries drives researchers to expand the charge cut-off voltage (COV) of cathode materials which can provide high specific capacity and

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discharge voltage [13–16].

A series of works have been reported in improving the COV of NCM materials. Gilbert et al. studied the cycle performance of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (NCM523)/graphite (Gr) cell, which underwent up to ~400 cycles in the voltage range of 3.00–4.40 V [17]. A linear capacity fading with cycle number which caused by the irreversible lithium loss in the solid electrolyte interphase (SEI) layer at anode was found. Chen et al. modified NCM523 to stabilize the host and interfacial structure of the material by a synergetic effect of element Zr and Ti. The specific capacities (3.00–4.40 V) achieve 163.9 mAh/g and 139.4 mAh/g at 1C and 16C, respectively [18]. The capacity retention after 200 cycles at 1C is 94.2%. The Zr/Ti co-modified NCM523 material shows excellent cycle and rate performance at high voltage. He et al. found that as an electrolyte additive tris(2,2,2-trifluoroethyl) phosphite (TTFP) could preferentially form a film on the surface of the cathode, thereby it can prevent oxidative breakdown of the electrolyte [19]. Hu et al. reported that by coating NCM523 with a uniform and thin layer of Li_2SiO_3 , the structure of NCM523 was stabilized and the lithium ion diffusion rate was greatly improved. The capacity retention as high as 96.1% was obtained for the Li_2SiO_3 -modified NCM523 after 100 cycles at 10C in a voltage range of 3.00–4.50 V [20].

However, the above investigations all relate to the NCM523 materials composed of aggregated sub-micrometer primary particles, formed from the agglomeration of secondary particles. This kind of material mostly in spherical structure and with low compaction density, is easy to broken and hence not suitable to improve the energy density of LIBs. Monocrystalline NCM has a similar structure with LiCoO_2 , but it possesses high compaction density and low grain boundary contact area, which is beneficial for the stability during long-term cycling especially under high voltage [21,22].

In this paper, 1100 mAh pouch lithium-ion cells with monocrystalline $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (MNCM523 for short) as cathode material and artificial graphite (AGr) as anode material are fabricated via a winding process. The electrochemical performances including specific capacities, rate capability, cycle life, HT< discharge capacities and storage of the cells at different COVs are studied. Electrochemical impedance spectroscopy (EIS) and in situ incremental capacity (ICA) are adopted to reveal the mechanism for capacity fading. The results provide a reference for development of high voltage NCM lithium-ion power batteries.

2. Experimental

2.1. Preparation of electrodes

The cathode slurry was prepared by mixing the active material (MNCM523, 96.0 wt%), carbon black (Super P, 1.5 wt%) and carbon nanotubes (CNT, 1.0 wt%) conductive additive, and polyvinylidene fluoride binder (PVDF, 1.5 wt%) dissolved in N-methyl pyrrolidone (NMP). The slurry was then coated on an aluminum foil (12 μm) with 28.8 mg/cm^2 active material loading and followed by drying at 110 °C for 12 h in vacuum oven (≤ -99.5 kPa, the same below), then the cathodes were cold pressed to a 98 μm thickness (the compact density was 3.60 g/cm^3) and were cut into a desired size.

The anodes was prepared by coating a slurry of mixed active material (AGr, 95.0 wt %), carbon black (Super P, 1.4 wt %) conductive additive, and carboxymethyl cellulose (CMC, 1.4 wt %) thickening agent and polymerized styrene butadiene rubber (SBR, 2.2 wt %) binder in distilled water (H_2O) on a copper foil (8 μm) with 16.8 mg/cm^2 active material loading and followed by drying at 110 °C for 12 h in vacuum oven. Then the anodes were cold pressed to a 122 μm thickness (the compact density was 1.55 g/cm^3) and were cut into a desired size.

2.2. Pouch lithium-ion cells fabrication

Electrode assembly, so-called as a jelly-roll was wound by winding machine with cathodes, ceramic polyethylene film (separator, 12 μm) and anodes. Then the jelly-roll was inserted into an aluminum plastic laminated film pouch, which was sealed by heat sealer to fabricate cell. The electrolyte is 1.0 M LiPF_6 dissolved in a mixed organic solvent consisting of ethyl methyl carbonate (EMC), ethylene carbonate (EC), diethyl carbonate (DEC), propylene carbonate (PC) in a volume ratio of 65:24:10:1 with additives of fluoro ethylene carbonate (FEC, 1.2 wt %), lithium difluoro (oxalate) borate (LiDFOB , 1.0 wt %) and lithium tetrafluoroborate (LiBF_4 , 0.5 wt %). Then a vacuum baking was adopted to control the moisture of cells into a relatively low level. After that, the electrolyte was injected into the cells in an Ar atmosphere glove box. The cells were pre-charged to 60% state of charge (SOC) at a constant current for formation in the charge and discharge equipment (NP-5AFF, Newpower, China) with a pressure of 310 kPa at 55 °C. After the initial formation, the cells were aged in a climate chamber for 24 h at 45 °C and then degassed to eliminate internal gas. Before testing, all the cells underwent two cycles with the constant current plus constant voltage (CC-CV) mode in a potential range of 3.00–4.35 V on the high precision battery test system (CT-4008-5V6A, Neware, China) at 25 °C to demonstrate reproducible capacity and resistance values. The 1C rated capacity of the cell was designed as 1100 mAh.

2.3. Electrochemical characterization

The charge-discharge capabilities, rate capability and cycle life were tested using the CT-4008-5V6A battery test system over different potential ranges: 3.0–4.30 V, 3.0–4.35 V and 3.0–4.40 V in CC-CV modes. Rate capability were measured at various rates (0.5C, 1C, 2C, 3C and 5C), respectively. HT< discharge and high temperature fully charged state storage tests were performed at 55 °C, –20 °C and 60 °C by an incubator (BETH-80, Bell, China), respectively. Long-term cycle life tests were conducted in the CC-CV mode with 1100 mA (1C) CC down to 22 mA (C/50) during the CV step. The incremental capacity analysis (ICA) was conducted at 0.04C CC charge-discharge, and the data was exported by an Agilent Data Acquisition/Switch Unit (34972A, Agilent, U.S). The EIS was conducted on an impedance analyzer (CS300, CorrTest, China) with frequency range from 1 MHz to 0.01 Hz with an A.C. amplitude of 5 mV. The measurements of fresh and cycled cells on state of half charged are both performed. In addition to HT< discharge and storage tests, all the above-described tests were conducted at room temperature. All the tests at different COVs are conducted on three cells with good reproducibility. Therefore, only the representative data of these tests are showed in this work.

3. Results and discussion

3.1. Structure and morphology of MNCM523 cathode

Fig. 1(a) displays the X-ray diffraction (XRD) pattern of the MNCM523 material. The result shows the characteristic diffraction peaks at 18.7°, 36.7°, 44.5° and 64.4°/65.0°, which correspond to the miller indices of (003), (101), (104) and (108)/(110), respectively. The pattern is indexed to isostructural with α - NaFeO_2 ascribed to the hexagonal R-3m space group [18] Fig. 1(b) exhibits the surface morphology of the MNCM523 material. It is seen that the material owns a smooth surface and a similar structure as a high-tap density LiCoO_2 [23] with a primary particle size between 1.50 and 8.00 μm . The microstructures indicate a typical monocrystalline structure of MNCM523 [8,24]. The element composition of the MNCM523 is tested according to the chemical titration and ICP offered by the

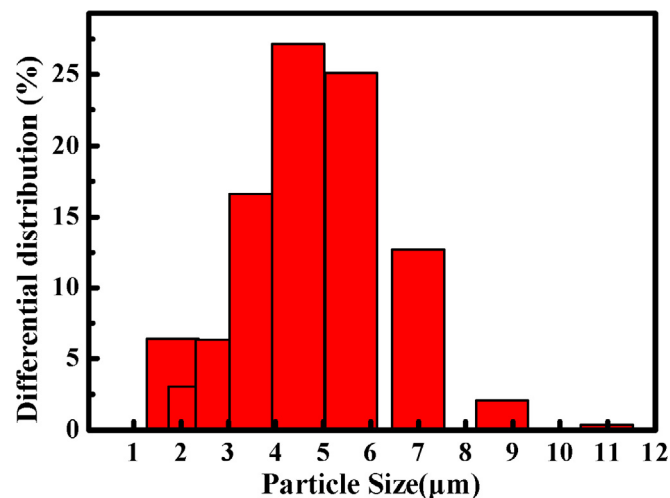
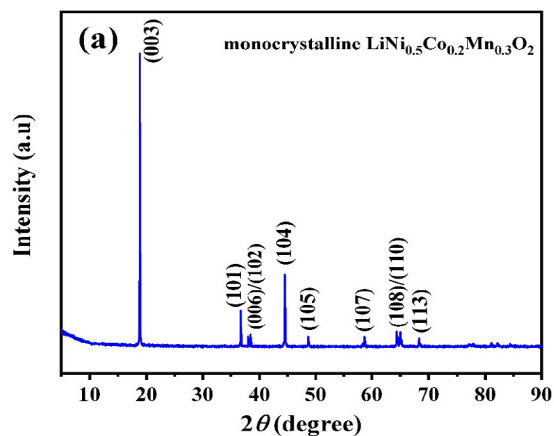


Fig. 2. Particle size distribution of monocrystalline $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$.

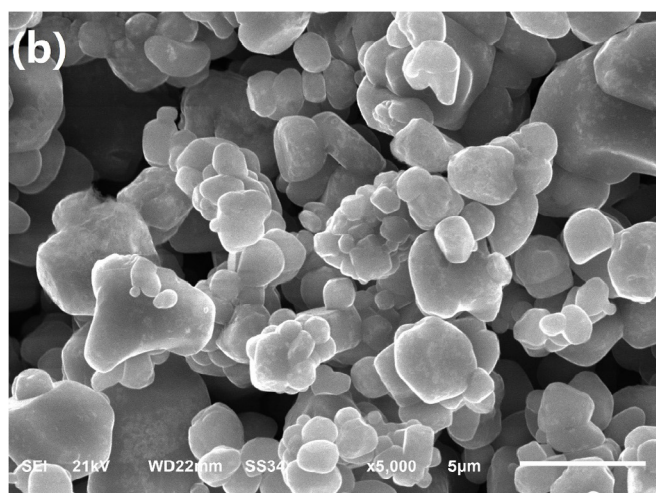


Fig. 1. XRD patterns (a) and SEM image (b) of monocrystalline $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ raw material.

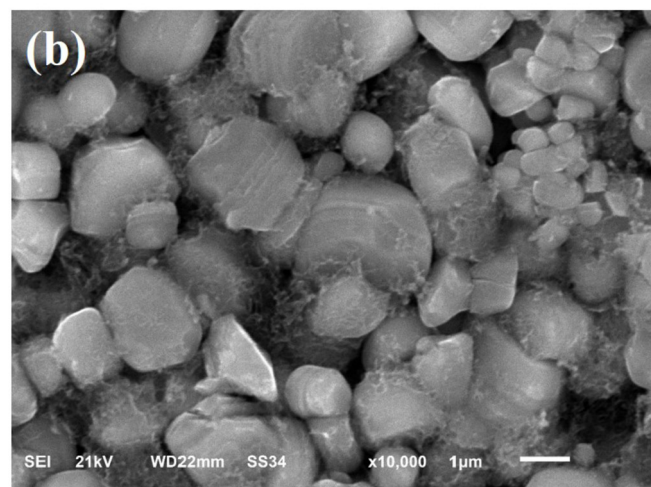
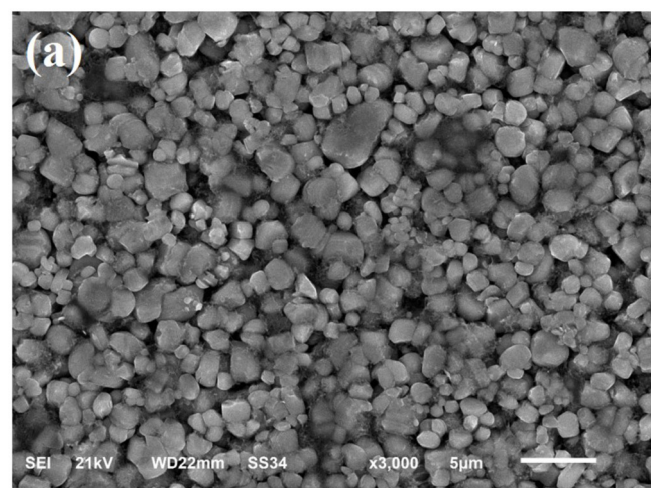


Fig. 3. SEM images of cathode electrode sheet with monocrystalline $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ after cold pressed.

material supplier. The contents of Li, Ni, Co and Mn are 13.4 wt%, 44.9 wt%, 17.7 wt% and 24.0 wt%, respectively, corresponding to a molar ratio of 12.9:5.1:2.0:2.9.

The particle size distribution of MNCM523 is shown in Fig. 2. It shows that MNCM523 has a uniform and concentrated particle distribution. The D50 (usually considered as the median particle diameter) value is 4.14 μm , which is smaller than that of the conventional NCM523 with particle sizes of 10.00–20.00 μm [18,20]. It is known that smaller particle size is beneficial to improve the transport rates of lithium ions at high voltage, the MNCM523 material exhibited a good electrochemical performance when applied as cathode material at high COV [25,26].

The morphology of MNCM523 cathode after cold pressing is displayed in Fig. 3. It is found that the conductive additives of CNT and SP are uniformly dispersed in the gaps of the particles and on the surface of the active material, which forms a good conductive structure [27]. Comparing with Fig. 1(b), a surface coating of conductive agents can be vaguely seen. The surface of the active material has no cracks, indicating a reasonable compaction density.

3.2. Electrochemical properties of the cells at different COVs

The discharge capacity, ohmic resistance, median voltage, specific capacity and energy density of the Li-ion pouch cell with

MNCM523 cathode and AGr anode at 1 C and different COVs are exhibited in Table 1. The specific capacity of MNCM523 at 3.0–4.4 V

Table 1
Electrochemical properties consisting of discharge capacity (mAh), ohmic resistance (ohm, mΩ), median voltage (V), specific capacity (mAh/g) and energy density (Wh/kg) of the cell at different COVs.

Item	COV(V)	Discharge Capacity(mAh)	ohm(mΩ)	Media Voltage(V)	Specific Capacity(mAh/g)	Energy Density (Wh/kg)
1#	4.30	1090	28.58	3.572	164.8	182.6
2#		1086	28.06	3.573	164.1	180.5
3#		1092	28.80	3.577	165.5	183.3
Geometric avg.		1089	28.48	3.574	164.80	182.1
1#	4.35	1141	28.90	3.585	172.6	192.4
2#		1143	29.12	3.585	172.7	193.2
3#		1143	29.20	3.587	173.5	195.1
Geometric avg.		1142	29.07	3.586	172.9	193.6
1#	4.40	1171	29.90	3.596	176.7	198.7
2#		1173	30.10	3.597	177.0	198.9
3#		1175	29.70	3.594	177.1	199.6
Geometric avg.		1173	29.90	3.596	176.9	199.1

is higher than the previous study [20]. According to the data in the table, as the COV increasing, the discharge capacity, median voltage, ohmic resistance, specific capacity and energy density of the cells are all improved in certain degrees. Fig. 4 shows more details about the specific capacity changes of the MNCM523 cathode at different COVs. The MNCM523 cathode presents higher specific capacity when the COVs is increased from 4.30 V to 4.40 V. The specific capacities increased from 4.30 V to 4.35 V and 4.35 V–4.40 V are 8.10 mAh/g (4.92%) and 4.00 mAh/g (2.30%), respectively. The former is more than twice higher than the latter. The enhancement of cathode specific capacity with the increase of COV is mainly because of the increased degree of Li^+ extraction in the cathode by the expanding external potential motivated.

Among them, the specific capacity improves greater when the COV increases from 4.30 V to 4.35 V than that from 4.35 V to 4.40 V. The possible reasons are considered as following: Firstly, higher cathode potential induces much severer electrolyte oxidation [28], and the side reactions are intensified; Secondly, the deeper degree of delithiation leads to lower instability of the cathode structure, and the $\text{Li}^+/\text{Ni}^{2+}$ mixing becomes severer serious [29]; Finally, the higher COV makes much more amount of Li^+ extracted from the cathode. While in the anode, there are no enough insert active sites for the insertion of Li^+ when the anode has the same active material loading density [30,31]. That means the higher COV needs a higher CB (capacity ratio of anode to cathode) value which could ensure

the Li^+ insertion. It is concluded that the increasing of the delithiation of MNCM523 becomes less effective with the further increase of COV. The specific capacity can only be improved in a reasonable extent according to the specific feature of certain cathode materials.

The media voltages, the discharge capacities and the energy densities of the pouch Li-ion full cell at 4.35 V and 4.40 V (Table 1) are all improved with the increase of COV. The increments of those values of the pouch cells nearly have the same trend as the variations of the specific capacity. While with the COV increasing, the ohmic resistance of the cells is increased gradually with the value below 1.0 mΩ.

3.3. Rate capability of the cell at different COVs

Fig. 5 presents the rate capability of the cell at different COVs. With the increase of the discharge rate, the discharge median voltages (Fig. 5(b)) and the capacity retention both decrease gradually (Fig. 5(a)). At the same rate, the capacity retention at the COV of 4.30 V is the highest, then is the retention at 4.35 V and the last of the retention at 4.40 V. The variations of the capacity retention at different COVs are consistent with the changes of the ohmic resistance as previously discussed. The increase in ohmic resistance results in larger ohmic voltage drop, and cell output voltage, the discharge median voltage as well as the capacity retention.

3.4. HT< performances of the cell at different COVs

In order to satisfy the application requirements at different regions with large temperature range, it is necessary to evaluate the thermal performances of LIBs at different discharge temperatures. Fig. 6 gives the discharge capacity retention at 1C and the kinetic impedances of the cell at different COVs at temperatures of -20°C (LT), 25°C (RT) and 55°C (HT), respectively. It is found that the COVs have relatively small effects on the discharge capacity retention at 1C at the different temperatures, which is probably due to the short discharging time and the different values of COV on the electrode-electrolyte interface. The discharge capacity of the COV at 4.30 V, 4.35 V and 4.40 V in HT are increased into 109.7%, 109.1% and 108.3%, respectively. The result indicates that the MNCM523 cathode has excellent high temperature performance, which is attributed to a fast lithium ion transport caused by the decreased viscosity of electrolyte and high electrode activity in HT. The high lithium ion transport rate and electrode activity also improve the kinetic performance of the electrodes and reduce the dynamic impedance of the cells significantly [32]. When at LT, the conductivity is reduced caused by the increase of electrolyte viscosity,

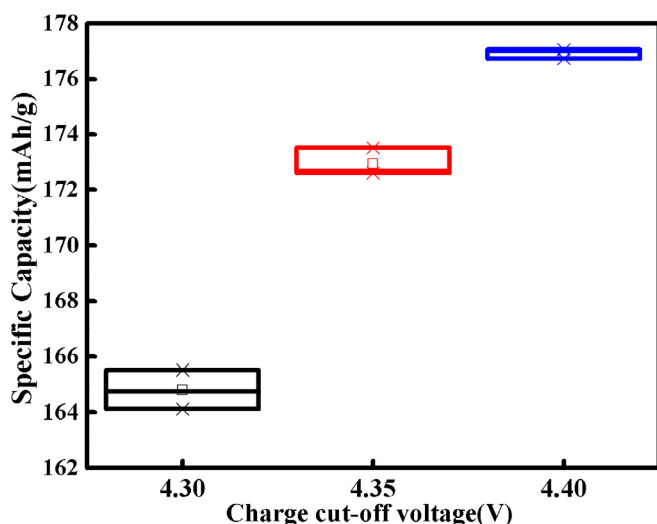


Fig. 4. Specific capacity comparison chart at different COVs.

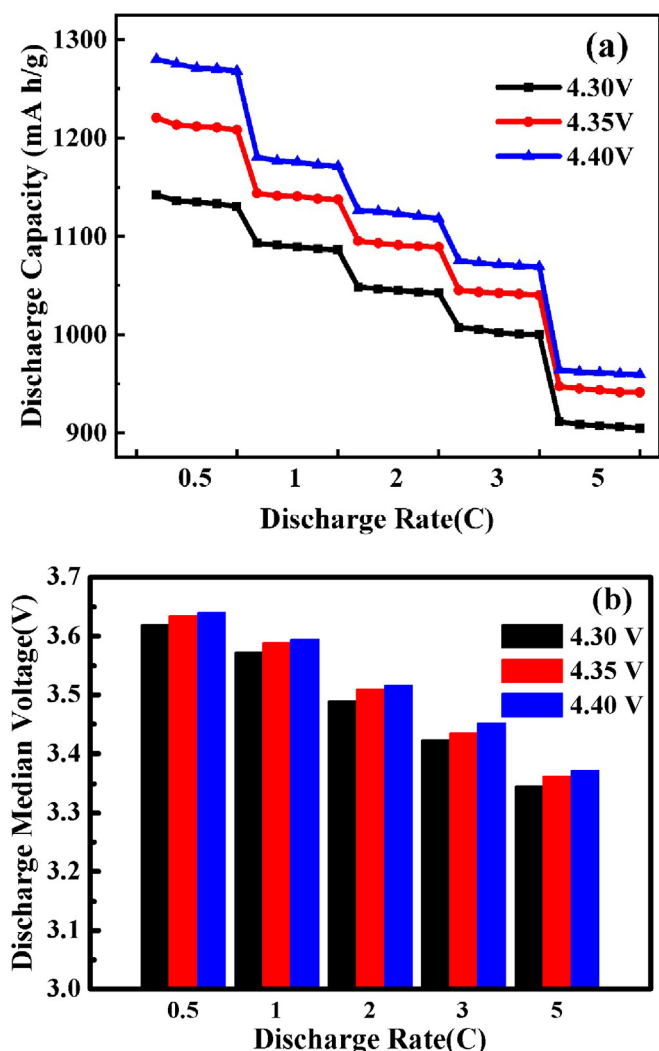


Fig. 5. Rate capability (a) and Discharge Median Voltage (b) of Rate capability at different COVs.

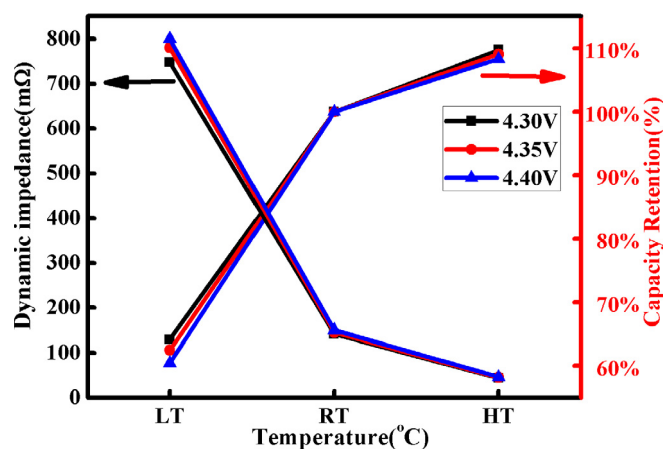


Fig. 6. Capacity retention (a) and Dynamic impedance (b) of HT/LT tests at different COVs.

electrode-electrolyte interfaces impedances and solid-phase diffusion resistances of Li^+ in both cathode and anode. All the above factors lead to the deterioration of the kinetic and

thermodynamic performances of electrodes with respect to the cell properties at room temperature [33,34]. Therefore, the impedance of the cells increases dramatically under LT with severe cell polarization, and the capacity retention is significantly lowered to 64.2%, 62.5%, and 60.4% at COVs of 4.30 V, 4.35 V and 4.40 V, respectively.

3.5. Storage performance of the cell at different COVs

Storage life is an important indicator to evaluate the long-term performance of Li-ion power batteries. High temperature storage accelerates the test and hence reduce the test time and expense. It is seen from Table 2 that the ohmic resistance after the storage increases gradually with the increase of COV, which is attributed to the increasing electrolyte oxidation on the cathode at the high potential and the facile electrolyte reduction on the anode at low potential. The oxidation and reduction of electrolyte promote the aggravation of parasitic side reactions and reduces the thicknesses of surface passivation layers [35,36]. In addition, the thickness of the cell increases as the increasing of COV. This is due to the thickened surface passivation layers. The decrease of OCV of the stored cell suggesting a more serious cells self-discharge. The capacity retention and recovery also decrease as the COV increasing, implying that both the reversible and irreversible lithium losses of the cells are intensified [37,38]. In short, the unwanted parasitic oxidation and reduction of electrolyte take place facilely with the COV increasing, thereby, the by-products are easily formed on the electrode surface which can block Li^+ transport and increase the overall cell impedance [39,40]. Meanwhile, parasitic side reactions consume active materials, in this case, the electrode cannot provide sufficient Li^+ insert active sites accordingly, which leads to an increase of an irreversible capacity loss and hence the reduction of discharge capacity. Therefore, it is necessary to improve the oxidant resistance of the electrolyte at the cathode and the reduction resistance ability at the anode as well as film-forming abilities at high COVs [28,41].

3.6. Cycle performance of the cell at different COVs

Cycle life is the most important factor for evaluating the long-term performance of Li-ion power batteries. Long cycle life means low usage cost, good environmental protection and reasonable utilization of resources. Fig. 7(a) shows the capacity retention of the as-prepared cells after 1500 cycles at 1 C under different COVs. Fig. 7(b)–(d) shows the discharge curves for selected cycles under COVs of 4.30 V, 4.35 V and 4.40 V, respectively. It is seen that the capacity retention within 800 cycles almost have the similar degradation rate and the values are all over 95.0%. However, with the further cycling, the cycle retention decreases much evidently with the increase of COVs. There are sudden drops of capacity for the different COVs, which could be caused by the growing electrolyte consumption, as reflected by the increase of interface impedance. After 1500 cycles, the retention of the cell at 4.30 V COV is still as high as 93.4%, while the values reduce to 87.4% and 80.3% for the COVs of 4.35 V at 4.40 V, respectively. The results indicate that as the COV increasing, the capacity retention of the MNCM523/AGr full cell gradually decreases.

Fig. 8 displays the Nyquist plots of the fresh and the cycled MNCM523/AGr full cells at different COVs. The insert is the corresponding equivalent circuit models. For the fresh cell, the Nyquist plot consists of a partially overlapped semicircle in high and intermediate frequency and a sloping line in low frequency. At high frequency, the intersection point on the real axis represents the ohmic resistance of the electrolyte, separator and electrodes, which is described as R_0 . The semicircle in high and intermediate frequency is attributed to the contributions from both electrodes and

Table 2
60 °C 14 d Storage results including variation of ohmic resistance (ohm, %) and thickness (T, %), retention of open-circuit voltage (OCV, %) and capacity (%), recover capacity (%) of the cell at different COVs.

Item	COV(V)	Variation of ohm (%)	Retention of OCV (%)	Variation of thickness (%)	Capacity retention (%)	Capacity recover (%)
1#	4.30	102.7	98.58	100.3	93.59	95.77
2#		102.2	98.74	100.3	94.98	96.30
3#		103.5	98.74	100.3	94.37	95.59
Geometric avg.		102.8	98.69	100.3	94.31	95.89
1#	4.35	107.9	97.36	100.3	92.23	94.14
2#		108.2	97.47	100.9	91.88	93.53
3#		107.8	97.43	101.3	91.62	92.76
Geometric avg.		108.0	97.42	100.8	91.91	93.47
1#	4.40	116.7	96.77	101.9	90.05	92.20
2#		115.3	97.21	101.6	89.92	91.89
3#		118.7	96.86	101.3	91.30	92.80
Geometric avg.		116.9	96.94	101.6	90.42	92.30

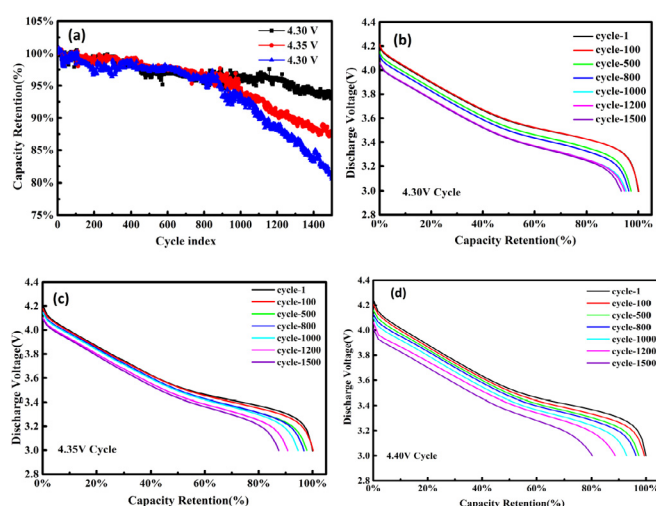


Fig. 7. Cycle performances of lithium-ion pouch cells with MNCM523 cathode at 1C under different COVs (a) and discharge curves at different cycles (1st, 100th, 500th, 800th, 1000th, 1200th and 1500th) under COV of 4.30 V(b), 4.35 V(c) and 4.40 V(d), respectively.

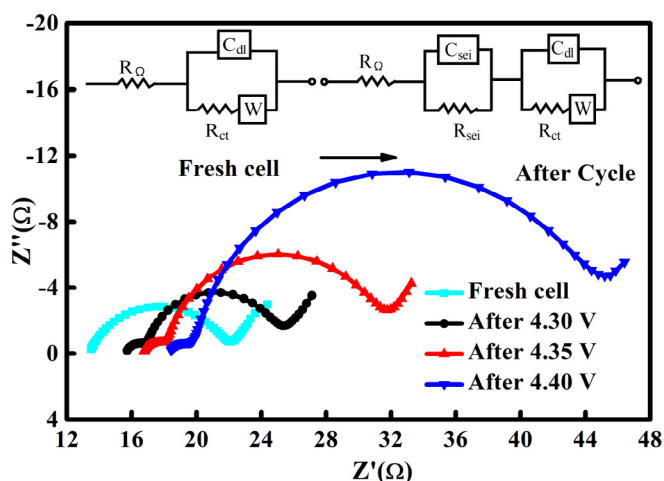


Fig. 8. Nyquist plots of the fresh and cycled cells at different COVs, the insert is the corresponding fitted equivalent electric circuit.

associated with charge transport across electrode-electrolyte interface (R_{ct}) and its relative double-layer capacitance (C_{dl}). The sloping line in low frequency corresponds to the Warburg

impedance (W) which reflects the Li^+ diffusion coefficient in the solid electrodes. However, after 1500 cycles at different COVs, the shape of the Nyquist plots changes totally and two semicircles are observed. The semicircle in high frequency is attributed to the surface films impedance (R_{sei}) and capacitance (C_{sei}) of two electrodes, and the semicircle in intermediate frequency corresponds to R_{ct} and C_{dl} [42,43]. Table 3 lists the data derived from the Nyquist plots of the fresh and cycled cells at different COVs. As can be seen from the figure and table, after aged for 1500 cycles, the R_Q of the cells at 4.30 V, 4.35 V and 4.40 V are increased into 15.65 Ω , 16.70 Ω , and 18.34 Ω , respectively. The differences of R_Q are 2.29 Ω , 3.34 Ω , and 4.98 Ω with respect to the value of the fresh cell, respectively. With the increase of COVs, the value of R_Q has a significant growth, which means that the contacts among active material, current collector, electrolyte and separator deteriorate. The R_{sei} of cycled cells are obvious increased, but they are only slightly increased after aged for 1500 cycles at different COVs. It can be speculated from the result that the passivating surface films of the electrodes are mainly formed during the charge/discharge preliminary process, and then are kept stable in the subsequent cycles. However, the difference of R_{ct} are significant, especially when at higher COVs. The R_{ct} values of the fresh cells at COVs of 4.30 V, 4.35 V and 4.40 V are calculated to be 8.25 Ω , 8.47 Ω , 13.05 Ω and 24.08 Ω , respectively, revealing a sharp increase of the charge-transfer resistance at high COVs. This is mainly induced by the degradation of the ionic pathways caused by chemical or physical changes at the surface films and electrodes as well as the block of electronic pathways caused by electrode-electrolyte interface changes. All the changes in the interface between electrode and electrolyte lead to the aggravation of the electrode kinetics [44]. The deterioration of the interface derived from the changes of EIS values is consistent with the cycle results of the cells in Fig. 7.

The degradation mechanism of capacity and power can be mainly classified to three main aspects, including the loss of active materials (LAM), the loss of lithium inventory (LLI), and the changes in reaction kinetics as often indicated by cell impedance increasing [45–47]. ICA analysis is a very sensitive, convenient, in situ and non-destructive analytical method, which can effectively distinguish degradation origins [48–50]. It is widely used in the analysis

Table 3
 R_Q , R_{sei} , R_{ct} values for the Nyquist plots of the fresh and cycled cell at different COVs.

Items	$R_Q(\Omega)$	$R_{sei}(\Omega)$	$R_{ct}(\Omega)$	$\Delta R_Q(\Omega)$	$\Delta R_{sei}(\Omega)$	$\Delta R_{ct}(\Omega)$
Fresh	13.36	—	8.252	0.000	—	0.000
4.30 V	15.65	1.335	8.467	2.290	1.335	0.215
4.35 V	16.70	1.495	13.05	3.340	1.495	4.798
4.40 V	18.34	1.558	24.80	4.980	1.558	16.55

of aging mechanisms such as cycle life and calendar life of battery. In the ICA curves, the peaks represent the process of phase transformation and the changes of the peak position indicate transformation of reaction kinetics which reflects in the changes of impedance. The peak intensity has a relationship with the amount of the active materials. That is, the varied amount of active lithium during certain processes can be quantified according to the peak areas [45,51]. Fig. 9 shows the ICA curves obtained from the fresh and cycled MNCM523/AGr full cells. For fresh cell, it can be seen from the figure that there are two distinct charging peaks at 3.47 V (1c) and 3.64 V (2c), corresponding to graphite lithium intercalation and MNCM523 delithiation phase transformation, respectively; and two distinct discharge peaks at 3.44 V (1d) and 3.61 V (2d) correspond to graphite delithiation and MNCM523 lithium intercalation phase transformation, respectively [39,40]. The voltage difference of lithiation/delithiation peaks between cathode and anode are both about 0.03 V, indicating that the cell has good reversibility. Nevertheless, both 2c and 2d peaks move slightly toward the high voltage direction after aged for 1500 cycles at different COVs, implying that the cathode impedance and the reversibility of the electrode change little. The apparent peaks indicate that the structures of the electrode remain stable even after 1500 cycles with high COVs. Nevertheless, the peak intensities and areas are diminished, representing LAM and LLI of the electrode. The LAM and LLI are more pronounced at 4.40 V, which could be caused by more amount of side reactions and more consumption of cathode active material induced by higher cathode potential. The insert is the lithiation (1c) and delithiation peaks (1d) of anode. As the COV increasing, the 1c and 1d peaks both move significantly towards the higher voltage direction, even almost disappears at 4.40 V. These results imply that the impedance and structural instability of the anode become large with the increase of COVs. Besides, the peak intensities and areas are also significantly weakened, which is caused by the LAM and LLI of the anode. The LAM and LLI also result in continuous deterioration of the electrode-electrolyte interfaces. In summary, as the increase of COV and aging cycle, the electrode-electrolyte interface deteriorates continuously, LAM increase and Li^+ insert active sites decrease. All of these lead to LLI, the aggravation of thermodynamic and kinetic performances as well as the grown of the surface film and charge transfer impedances. Then, the capacity fading aggravates and the cell life shortens.

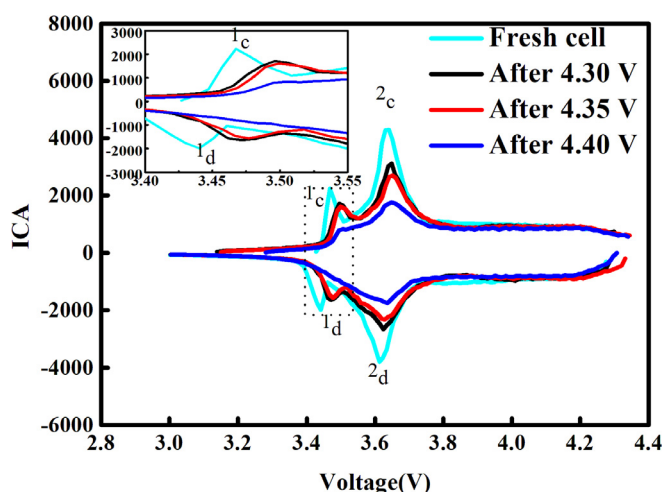


Fig. 9. The ICA plots for the fresh and cycled cells at different COVs, inset is the curves at enlarged region of 3.40–3.55 V.

4. Conclusions

In this report, the electrochemical performance of a monocrystalline $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ /artificial graphite pouch Li-ion full cell charged to different cut-off voltages is studied. The results show that the specific capacities of MNCM523 are 164.8 mAh/g, 172.9 mAh/g and 176.9 mAh/g at 4.30 V, 4.35 V and 4.40 V, respectively. The discharge capacity, median voltage and energy density of the cells all improved with the increase of COV. The ohmic resistance also increases simultaneously, and the rate capability, HT< discharge capacity and storage performance are slightly worse. The cycle performances of the cell at different charge cut-off voltages become obvious different after 1000 cycles. The capacity fading is also aggravated as the COV increasing. The capacity retention at 4.30 V, 4.35 V and 4.40 V after 1500 cycles are 93.4%, 87.4% and 80.3%, respectively. The EIS and ICA tests of fresh and cycled cells indicate that the main reasons of capacity fading are the continuous LLI and LAM due to the deterioration of the electrode-electrolyte interfaces, and the increase of charge transfer resistance. Therefore, stabilizing the electrode-electrolyte interfaces, improving the stability of the electrolyte and reducing the interface impedance are the main factors to improve the performances the monocrystalline NCM Li-ion power battery at high charge cut-off voltage.

Acknowledgement

The authors are grateful for the financial supports from the Public Projects of Zhejiang Province (Grant No. LGG19E020001). The authors are also grateful for the monocrystalline $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ material supports from Ningbo Ronbay Technology Lithium Battery Material Co., Ltd (No.39, East Road of Tanjialong, Yuyao, China), the electrolytes from Guangzhou Tinci Materials Technology Co., Ltd. (No.8, Kangda Road, Yunpu Industrial Zone, Huangpu Region, Guangzhou, China) and Zhuhai Smoothway Electronic Materials Co., Ltd. (No.2, Nanhua Road of South Town, Gaolan Port Economic Zone, Zhuhai, China), and et al.

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